

STButton: Exploring Opportunities for Buttons with Spatio-Temporal Tactile Output

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ABSTRACT

We present STButton, a physical button with a high-resolution spatio-temporal tactile feedback surface. The 5 x 8 pin array tactile display size of 20mm x 28mm enables buttons to express various types of information, such as value with the number of raised pins, direction with the location of raised pins, and duration of time with blinking animation. With a highly expressive tactile surface, the button can seamlessly transfer assistive feedforward and feedback during spontaneous button interaction, such as touching to locate the button or applying gradual pressure to press the button. In the demonstration, attendees experience five scenarios of button interaction: the seat heater button on a car, the volume control button on a remote controller, the power button on a laptop, the menu button on a VR controller, and the play button on a game controller. In each scenario, the representative role of tactile feedback is configured differently, allowing attendees to experience the rich interaction space and potential benefits of STButton. Early accessed attendees appreciated the unique opportunity to transfer information with a highly expressive tactile surface and emphasized that STButton adds a tangible layer to the user experience, enhancing emotional and sensory engagement.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**.

KEYWORDS

Haptic, Haptic button, Physical button, Spatio temporal tactile output

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1 INTRODUCTION

Physical buttons have long served as the fundamental means of interacting with various daily devices. Their protruding physical form and distinctive 'bottom-out' click sensation tactically provide a useful assistant, offering tangible feedback on button placement and confirmation of input. The tactile channel of a button can seamlessly provide various assistive information while users touch and press buttons for interaction.

In the exploration of new avenues in physical button interaction, recent research has delved into leveraging the tactile channel with additional tactile actuators. Park et al. [12, 13] explored the perception dimension and information transmission capacity of vibrotactor-augmented physical buttons. Through potential use scenarios, the researchers suggested that the tactile channel can assist button interaction, such as providing immediate interaction result feedback with vibrotactile sensations upon button press.

Inspired by recent research, we aim to further explore the potential interaction space of the highly expressive tactile channel button. Specifically, we would like to open a discussion: what if button tactile channels could convey more detailed information, such as system values like the current volume and user input status like long-pressing duration? What if button tactile channels deliver information not only instantaneously at button press but also during touching to locate the button, or even while the finger rests on the button?

To expand the tactile expressiveness of the button further, we implemented STButton, a button device with a high-resolution spatio-temporal tactile surface. In this interactivity, we demonstrate six design probes using STButton to initiate a discussion about the possibilities of utilizing high-resolution tactile feedback on buttons.

2 STBUTTON SYSTEM

The STButton is a mechanically-pressable button device designed to be pressed up to 0.5cm by a finger. To provide high resolution tactile feedback on the pressing finger, the button's surface was designed to resemble a braille display of five by eight pins. We used four braille modules from Metec to build this display, specifically the Modul D2. Each braille module houses five by two pins with dot spacing of 2.5mm, dot stroke of 0.7mm, and 0.3N tactile force. Pins are operated by piezo-benders, and dot rising time is clocked at 100ms. The size of the entire module is 12.5x5x51.8mm. To control the



Figure 1: STButton 3D model and prototype

module, we utilized Arduino Pro, connecting the module using SPI. Modules are powered by a dc-hvdc converter (A02P-5, XP Power). Beyond the tactile display, we incorporated a linear magnetic sensor to detect the displacement of the button. Furthermore, an Inertial Measurement Unit (IMU) was integrated, enabling the system to discern the button's orientation.

Rationale Behind Spatio-Temporal Tactile Feedback Selection. Our primary objective was to delve into the concept of a button that could offer highly expressive tactile feedback. In tactile feedback, spatio-temporal haptic devices stand out for their superior expressiveness. They offer users the capability to discern object shapes through delineated contours [10], perceive distinct temporal patterns [8], pinpoint specific object locations [7], provide intuitive directional cues [9, 15], and even replicate the sensation of diverse textures [2].

Rationale Behind Braille Display Module Selection. From various types of spatio-temporal tactile actuators[3–6, 9, 11], we targeted to choose one with a spatial resolution closely mimicking the tactile spatial threshold of the fingertip. It was also important that the module had a fast enough update rate not to impede button interaction. In light of these requirements, D3 Modul from Metec was deemed ideal, as it satisfied crucial prototyping requisites, including the desired spatial resolution and operational speed.

Rationale for Button Dimensions. Drawing upon the findings of Alexander et al. [1], an average button typically spans a width of approximately 18mm. With this benchmark in mind, our design aim was to make a button sizable enough to accommodate both the thumb and the index fingertip areas comfortably. The core of the button measures 20 mm in width and 28 mm in length. Within this space, a 15 mm by 23 mm area is designated for the braille display.

2.1 Tactile Expressivity of the STButton

Grounding on the work of Pietrzak et al.[14], braille display can generate static and dynamic patterns. The static pattern can be defined as a single spatial pattern, and the single spatial pattern is a combination of shape, location, and size. The dynamic pattern is a series of static patterns.

Fig. 2 are examples of static and dynamic tactile patterns that can be made with shape, location, and size combinations. As a static pattern, the location of the activated pins can indicate the value (Fig. 2(a)). Line location can be used to indicate the system value on volume buttons that change the system value by button press. As

the user presses the button, the system value increases, and so does the location of the line. As a static pattern, the size of the activated pins can indicate the level (Fig. 2(b)). Trigger buttons on the game controller allow the user to manipulate the input level with input force, and rectangle size can indicate the input level. As the user gradually increases the pressure on the button, the input level and the rectangle size increase. As a static pattern, the shape of the activated pins can indicate the semantic value (Fig. 2(c)). Letter buttons on the keyboard can change their tactile iconic shape as the user presses the modifier key. As a dynamic pattern, blink can be used as an affordance, inviting users to press a button (Fig. 2(d)). As a dynamic pattern, a cycle of rotation can express the duration of time (Fig. 2(e)). On buttons that need long presses, the dot rotation pattern indicates how long the user has pressed the button.

3 STBUTTON DEMOS

We demonstrate five STButton working demos in which highly expressive tactile feedback is essential in providing feedforward or feedback. Each demo consists of experiencing one fully functioning button usage scenario. Five demos are controlling the seat heat level, controlling speaker volume, force rebooting the laptop, executing the menu on the VR device, and playing and restarting the game. Scenarios are selected to show the following potential role of tactile feedback: feedforward the system status before button pressing (Heat seat button), feedback the updated system status during and after button pressing (Volume control button), feedback the user input during button pressing (Power button), feedforward the user input during button pressing (VR menu button) and signifying button press (Game controller button).

Demo applications are implemented as web applications using Flask, and we use the laptop to display the simulated button-using environment. The STButton prototype is connected to the laptop for controlling the simulated environment.

3.1 Demo 1: In-car Heated Seat Button

The user can perceive the current heating level by touching the constant tactile pattern on the STButton even without looking at it; the driver's eyes can always be on the road. In this scenario, the participants saw a photo of a driver's view on the screen in front, with three bars in the bottom right corner visualizing the heating level. The participants used their right index finger to press the STButton fixed on the desk, assuming the driver's seat on the left. Each press increased the heating level, and another press at the

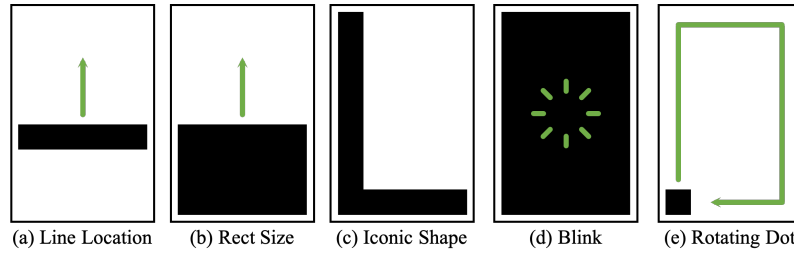


Figure 2: STButton tactile pattern examples

maximum level turned the heating off. The tactile pattern showed the current level of heating out of four, including the off-state at the beginning. From no heat to maximum heat, each level corresponded to tactile patterns of zero, three, five, and eight (all) rows of pins activated from the bottom, offering tactile feedback consistent with onscreen visual representation. The entire pins went down in the press-down state to acknowledge the pressing action. Then, only corresponding pins were activated in the release-press state, and they remained raised after the disengaged state.

3.2 Demo 2: Volume Control Button on a Remote Controller

The user can recognize the current volume level by tactile, which also prevents cases such as accidentally playing the video at unwantedly loud volumes in quiet public places. The participants saw an image of a room with speakers and held the prototype in their right hand as if holding a remote controller. With classical music constantly being played, they pressed the button with their thumb to adjust the volume out of 41 distinct levels. Initially starting at level 12, a brief press in/decreased the volume by one level, and a continuous hold in/decreased the volume until the release-press state. We added a 2-second buffer before the hold mode initiates to avoid unintended manipulation. The tactile pattern was designed to express each level differently by accumulatively activating pins one by one from the left bottom corner, row by row, and left to right. No pins represented zero volume, and all pins indicated a maximum volume of 40. Since we had one prototype, the participants were demonstrated with volume increase and decrease modes separately upon request.

3.3 Demo 3: Laptop Power Button

As we force reboot a laptop after it freezes, it is sometimes difficult to tell whether it is responding. In this scenario, the user can receive temporal tactile feedback on the current system status. The participants saw a photo of a laptop and were asked to press the prototype fixed on the table with their right index finger. If the button was pressed for over three seconds, the blink pattern with a fastening pace was played to indicate that the force reboot timer was filling up. Then, the double-blink pattern followed to indicate that a force reboot was initiated. After the reboot was done, namely seven seconds after activation in this scenario, another double-blink pattern was provided, no matter whether the participant still had one's finger on the button.

3.4 Demo 4: VR Radial Menu Select Button

People perform various item selection tasks in VR scenarios, but sometimes, using a ray from the VR controller to precisely point to the target item is inconvenient or restricted. This scenario provides tactile feedback on which menu item is selected from a four-way radial menu. It enables four distinct functioning buttons, or possibly even more, to be combined into a single STButton instead of exactly pointing to small buttons. We provided participants with an image of the space to represent the VR environment, and they held the prototype like a VR controller in their right hand to press the button with their thumb. From the position of the press-down, one can tilt the STButton up, down, left, or right for more than five degrees to select from the four items. The currently selected item is also displayed as corresponding tactile feedback consisting of two columns (left/right) or three rows (top/bottom) activated along the border of the tactile display. The selected item at the release-press state was regarded as the final selection.

3.5 Demo 5: Game Controller Button

The STButton prototype itself can act as a multipurpose game controller button, and we demonstrated the scenario with Google's Dinosaur Game. The participants were asked to hold the prototype in their right hand and press the button with their thumb. We modified the game so that the dinosaur does three levels of jumps to avoid obstacles triggered by the depth of the press. Tactile feedback of three, five, and eight (all) rows activated from the bottom was provided accordingly to provide the participants with a hint of the current depth pressed, or the height of jump to be triggered, intuitively at press-down. A single press was used to initiate the game, and as the dinosaur collided with the obstacle, a double press was used to restart the game. At this point, we provided periodic double-blink patterns to hint the users about the action to do next. All such functionalities were meant to show the adaptability of the STButton based on various contextual demands, serving as an auxiliary component to enhance the overall gaming experience.

4 CONCLUSION

In the continually evolving landscape of human-computer interaction, traditional input mechanisms, like buttons, remain pivotal. However, their typical passive nature in interactions has often limited their potential. We demonstrated the STButton button with a high-resolution spatio-temporal tactile surface. We aim to discuss the possibilities of leveraging high-resolution tactile channels

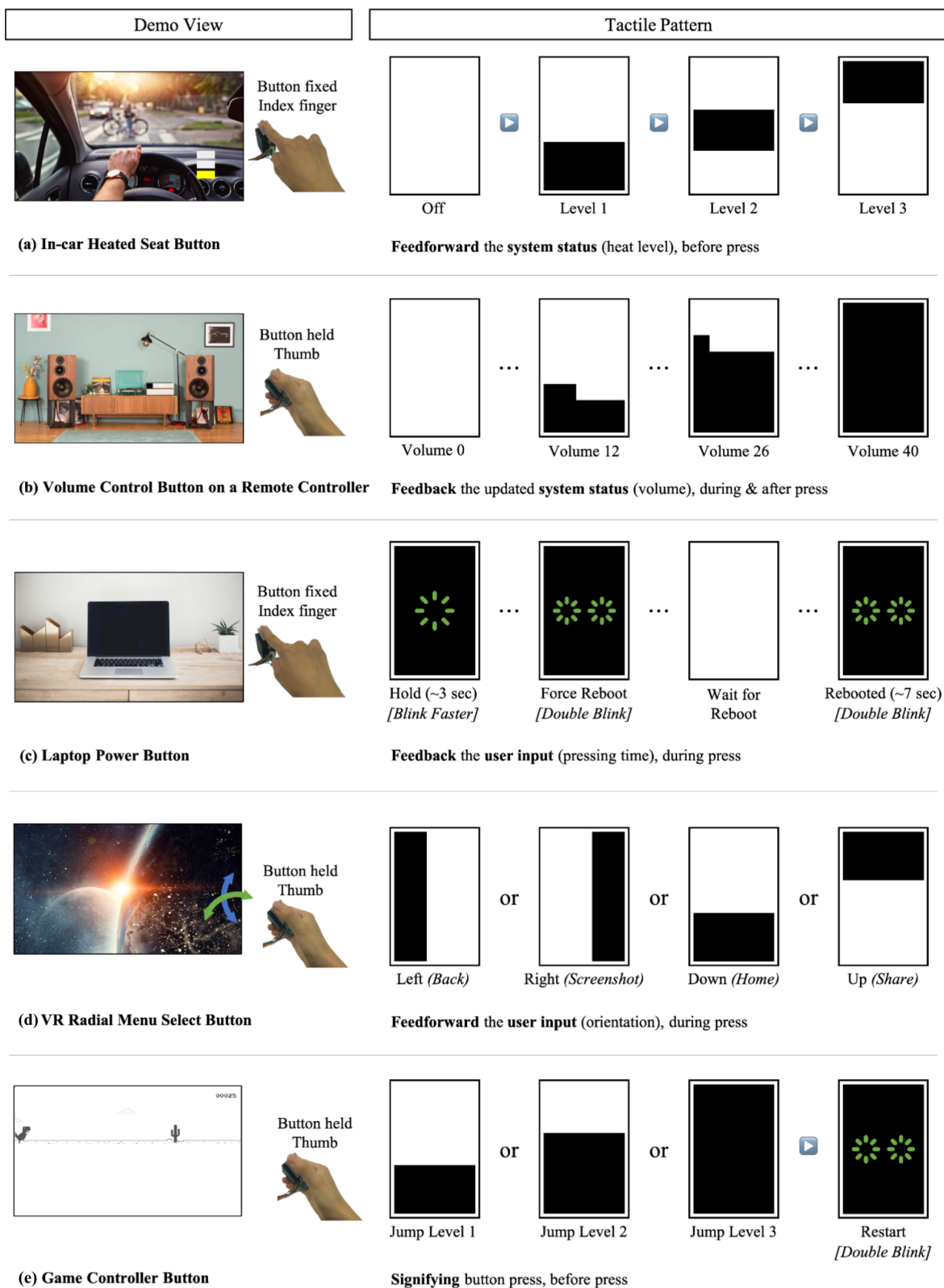


Figure 3: Demos

for enhancing button interaction. We presented five possible interaction scenarios utilizing tactile feedback. The early feedback from users supports that STButton emerged as a valuable tool for improving user experience by providing immediate and intuitive feedback. We expect this demo to open up new discussions from unique characteristics of spatio-temporal tactile haptics on button interaction and new opportunities for the button.

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